

Task :-- having, groupby and order by, limit, asc/desc

	product_id	product_name	category	price	rating	stock	brand	sold_quantity
▶	11	echo dot	electronics	3499.00	4.6	120	amazon	500
	12	kindle paperwhite	electronics	9999.00	4.8	80	amazon	300
	13	fire tv stick	electronics	3999.00	4.5	50	amazon	450
	14	instant pot duo	kitchen	7999.00	4.7	30	instantpot	200
	15	air fryer	kitchen	5999.00	4.6	40	philips	150
	16	samsung galaxy s23	mobiles	69999.00	4.8	25	samsung	100
	17	oneplus 11	mobiles	49999.00	4.7	20	oneplus	90
	18	bose headphones	electronics	19999.00	4.5	15	bose	80
	19	logitech mouse	electronics	1299.00	4.4	200	logitech	600
	20	amazon basics bag	accessories	999.00	4.3	300	amazon	400
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

5 Questions on GROUP BY

1. Find the **total sold_quantity** of products for each category.

```
select category,sum(sold_quantity) as totalQuantity
```

```
from products
```

```
group by category;
```

	category	totalQuantity
▶	electronics	1930
	kitchen	350
	mobiles	190
	accessories	400

2. Find the **average price** of products for each brand.

```
select brand,avg(price) as avgPrice
```

```
from products
```

```
group by brand;
```

	brand	avgPrice
▶	amazon	4624.000000
	instantpot	7999.000000
	philips	5999.000000
	samsung	69999.000000
	oneplus	49999.000000
	bose	19999.000000
	logitech	1299.000000

3. Count how many products are available in each category.

```
select category,count(product_id) as countProducts
from products
group by category;
```

	category	countProducts
▶	electronics	5
	kitchen	2
	mobiles	2
	accessories	1

4. Find the **maximum rating** of products for each category.

```
select category,max(rating) as maximumRating
from products
group by category;
```

	category	maximumRating
▶	electronics	4.8
	kitchen	4.7
	mobiles	4.8
	accessories	4.3

5. Find the **minimum price** of products for each brand.

```
select brand,min(price) as minimumPrice
from products
group by brand;
```

	brand	minimumPrice
▶	amazon	999.00
	instantpot	7999.00
	philips	5999.00
	samsung	69999.00
	oneplus	49999.00
	bose	19999.00
	logitech	1299.00

5 Questions on HAVING

- Find all categories where the **total sold_quantity** is greater than 1000.

select category, sum(sold_quantity) as totalQuantity

from products

group by category

having totalQuantity >1000;

Result Grid

↺↻

Filter Rows:

	category	totalQuantity
▶	electronics	1240

- Find brands where the **average price** of products is more than 10000.

select brand, avg(price) as avgPrice

from products

group by brand

having avgPrice >10000;

Result Grid




Filter Rows:

	brand	avgPrice
▶	asus	58999.000000
	lenovo	63999.000000
	jbl	11999.000000

- Find categories where the **maximum rating** is greater than 4.7.

select category ,max(rating) as maxRating

from products

group by category

having maxRating >4.7;

Result Grid	Filter Rows
category	maxRating

4. Find brands where the **count of products** is more than 2.

select brand, count(sold_quantity) as productCount

from products

group by brand

having productCount>2;

Result Grid	Filter Rows
brand	productCount
▶ amazon	3

5. Find categories where the **sum of stock** is less than 200.

select category, sum(stock) as totalStock

from products

group by category

having totalStock <200;

Result Grid	Filter Rows
category	totalStock
▶ computers	65
fashion	160
wearables	120

5 Questions on ORDER BY

1. List all products **ordered by price ascending**.

select *

from products

order by price asc;

	product_id	product_name	category	price	rating	stock	brand	sold_quantity
▶	20	amazon basics bag	accessories	999.00	4.3	300	amazon	400
	19	logitech mouse	electronics	1299.00	4.4	200	logitech	600
	11	echo dot	electronics	3499.00	4.6	120	amazon	500
	13	fire tv stick	electronics	3999.00	4.5	50	amazon	450
	15	air fryer	kitchen	5999.00	4.6	40	philips	150
	14	instant pot duo	kitchen	7999.00	4.7	30	instantpot	200
	12	kindle paperwhite	electronics	9999.00	4.8	80	amazon	300
	18	bose headphones	electronics	19999.00	4.5	15	bose	80
	17	oneplus 11	mobiles	49999.00	4.7	20	oneplus	90
	16	samsung galaxy s23	mobiles	69999.00	4.8	25	samsung	100
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

2. Show all products **ordered by sold_quantity descending**.

	product_id	product_name	category	price	rating	stock	brand	sold_quantity
▶	19	logitech mouse	electronics	1299.00	4.4	200	logitech	600
	11	echo dot	electronics	3499.00	4.6	120	amazon	500
	13	fire tv stick	electronics	3999.00	4.5	50	amazon	450
	20	amazon basics bag	accessories	999.00	4.3	300	amazon	400
	12	kindle paperwhite	electronics	9999.00	4.8	80	amazon	300
	14	instant pot duo	kitchen	7999.00	4.7	30	instantpot	200
	15	air fryer	kitchen	5999.00	4.6	40	philips	150
	16	samsung galaxy s23	mobiles	69999.00	4.8	25	samsung	100
	17	oneplus 11	mobiles	49999.00	4.7	20	oneplus	90
	18	bose headphones	electronics	19999.00	4.5	15	bose	80
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. Display products **ordered by rating descending and then price ascending**.

select *

from products

order by rating desc,price asc;

	product_id	product_name	category	price	rating	stock	brand	sold_quantity
▶	12	kindle paperwhite	electronics	9999.00	4.8	80	amazon	300
	16	samsung galaxy s23	mobiles	69999.00	4.8	25	samsung	100
	14	instant pot duo	kitchen	7999.00	4.7	30	instantpot	200
	17	oneplus 11	mobiles	49999.00	4.7	20	oneplus	90
	11	echo dot	electronics	3499.00	4.6	120	amazon	500
	15	air fryer	kitchen	5999.00	4.6	40	philips	150
	13	fire tv stick	electronics	3999.00	4.5	50	amazon	450
	18	bose headphones	electronics	19999.00	4.5	15	bose	80
	19	logitech mouse	electronics	1299.00	4.4	200	logitech	600
	20	amazon basics bag	accessories	999.00	4.3	300	amazon	400
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

4. List all products **ordered by category ascending and sold_quantity descending**.

select *

from products

```
order by sold_quantity desc, category asc;
```

[illegible]

5. Show all products **ordered by brand alphabetically**.

```
select *
```

from products

```
order by brand asc;
```

[illegible]

5 Questions on LIMIT

1. Show the **first 5 products** in the table.

```
select *
```

from products

```
limit 5;
```

[illegible]

2. Show the **top 3 highest-priced products**.

select *

from products

order by price desc

limit 3;

	product_id	product_name	category	price	rating	stock	brand	sold_quantity
▶	16	samsung galaxy s23	mobiles	69999.00	4.8	25	samsung	100
	17	oneplus 11	mobiles	49999.00	4.7	20	oneplus	90
	18	bose headphones	electronics	19999.00	4.5	15	bose	80
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. Show the **first 5 products when ordered by rating descending**.

select *

from products

order by rating desc

limit 5;

	product_id	product_name	category	price	rating	stock	brand	sold_quantity
▶	12	kindle paperwhite	electronics	9999.00	4.8	80	amazon	300
	16	samsung galaxy s23	mobiles	69999.00	4.8	25	samsung	100
	14	instant pot duo	kitchen	7999.00	4.7	30	instantpot	200
	17	oneplus 11	mobiles	49999.00	4.7	20	oneplus	90
	11	echo dot	electronics	3499.00	4.6	120	amazon	500
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

4. Show **2 products after skipping the first 5 rows**.

select * from products

limit 2 offset 5;

	product_id	product_name	category	price	rating	stock	brand	sold_quantity
▶	16	samsung galaxy s23	mobiles	69999.00	4.8	25	samsung	100
	17	oneplus 11	mobiles	49999.00	4.7	20	oneplus	90
•	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

5. Display the **top 10 products with highest sold_quantity**.

select *

from products

order by sold_quantity desc

limit 10;

	product_id	product_name	category	price	rating	stock	brand	sold_quantity
▶	19	logitech mouse	electronics	1299.00	4.4	200	logitech	600
	11	echo dot	electronics	3499.00	4.6	120	amazon	500
	13	fire tv stick	electronics	3999.00	4.5	50	amazon	450
	20	amazon basics bag	accessories	999.00	4.3	300	amazon	400
	12	kindle paperwhite	electronics	9999.00	4.8	80	amazon	300
	14	instant pot duo	kitchen	7999.00	4.7	30	instantpot	200
	15	air fryer	kitchen	5999.00	4.6	40	philips	150
	16	samsung galaxy s23	mobiles	69999.00	4.8	25	samsung	100
	17	oneplus 11	mobiles	49999.00	4.7	20	oneplus	90
	18	bose headphones	electronics	19999.00	4.5	15	bose	80
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

5 Questions on ASC / DESC

1. Show all products **sorted by price ascending**.

	product_id	product_name	category	price	rating	stock	brand	sold_quantity
▶	20	amazon basics bag	accessories	999.00	4.3	300	amazon	400
	19	logitech mouse	electronics	1299.00	4.4	200	logitech	600
	11	echo dot	electronics	3499.00	4.6	120	amazon	500
	13	fire tv stick	electronics	3999.00	4.5	50	amazon	450
	15	air fryer	kitchen	5999.00	4.6	40	philips	150
	14	instant pot duo	kitchen	7999.00	4.7	30	instantpot	200
	12	kindle paperwhite	electronics	9999.00	4.8	80	amazon	300
	18	bose headphones	electronics	19999.00	4.5	15	bose	80
	17	oneplus 11	mobiles	49999.00	4.7	20	oneplus	90
	16	samsung galaxy s23	mobiles	69999.00	4.8	25	samsung	100
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

2. Show all products **sorted by rating descending**.
3. Display products **sorted by sold_quantity descending, then price ascending**.

	product_id	product_name	category	price	rating	stock	brand	sold_quantity
▶	19	logitech mouse	electronics	1299.00	4.4	200	logitech	600
	11	echo dot	electronics	3499.00	4.6	120	amazon	500
	13	fire tv stick	electronics	3999.00	4.5	50	amazon	450
	20	amazon basics bag	accessories	999.00	4.3	300	amazon	400
	12	kindle paperwhite	electronics	9999.00	4.8	80	amazon	300
	14	instant pot duo	kitchen	7999.00	4.7	30	instantpot	200
	15	air fryer	kitchen	5999.00	4.6	40	philips	150
	16	samsung galaxy s23	mobiles	69999.00	4.8	25	samsung	100
	17	oneplus 11	mobiles	49999.00	4.7	20	oneplus	90
	18	bose headphones	electronics	19999.00	4.5	15	bose	80
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

4. List products **sorted by category ascending, then sold_quantity descending**.

	product_id	product_name	category	price	rating	stock	brand	sold_quantity
▶	20	amazon basics bag	accessories	999.00	4.3	300	amazon	400
	19	logitech mouse	electronics	1299.00	4.4	200	logitech	600
	11	echo dot	electronics	3499.00	4.6	120	amazon	500
	13	fire tv stick	electronics	3999.00	4.5	50	amazon	450
	12	kindle paperwhite	electronics	9999.00	4.8	80	amazon	300
	18	bose headphones	electronics	19999.00	4.5	15	bose	80
	14	instant pot duo	kitchen	7999.00	4.7	30	instantpot	200
	15	air fryer	kitchen	5999.00	4.6	40	philips	150
	16	samsung galaxy s23	mobiles	69999.00	4.8	25	samsung	100
	17	oneplus 11	mobiles	49999.00	4.7	20	oneplus	90
✱	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

5. Show all products **sorted by brand descending**.

	product_id	product_name	category	price	rating	stock	brand	sold_quantity
▶	16	samsung galaxy s23	mobiles	69999.00	4.8	25	samsung	100
	15	air fryer	kitchen	5999.00	4.6	40	philips	150
	17	oneplus 11	mobiles	49999.00	4.7	20	oneplus	90
	19	logitech mouse	electronics	1299.00	4.4	200	logitech	600
	14	instant pot duo	kitchen	7999.00	4.7	30	instantpot	200
	18	bose headphones	electronics	19999.00	4.5	15	bose	80
	11	echo dot	electronics	3499.00	4.6	120	amazon	500
	12	kindle paperwhite	electronics	9999.00	4.8	80	amazon	300
	13	fire tv stick	electronics	3999.00	4.5	50	amazon	450
	20	amazon basics bag	accessories	999.00	4.3	300	amazon	400
✱	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

	product_id	product_name	category	price	rating	stock	brand	sold_quantity
1		Echo Dot	Electronics	3499	4.6	120	Amazon	500
2		Kindle Paperwhite	Electronics	9999	4.8	80	Amazon	300
3		Fire TV Stick	Electronics	3999	4.5	50	Amazon	450
4		Instant Pot Duo	Kitchen	7999	4.7	30	InstantPot	200
5		Air Fryer	Kitchen	5999	4.6	40	Philips	150
6		Samsung Galaxy S23	Mobiles	69999	4.8	25	Samsung	100
7		OnePlus 11	Mobiles	49999	4.7	20	OnePlus	90
8		Bose Headphones	Electronics	19999	4.5	15	Bose	80
9		Logitech Mouse	Electronics	1299	4.4	200	Logitech	600
10		Amazon Basics Bag	Accessories	999	4.3	300	Amazon	400